Week 4 - Learning Map

Goal:

Understand how to design and build multi-agent systems using LangGraph, coordinate specialized agents through graph workflows, apply subagent design principles, and implement a travel planning system that integrates multiple agents to produce unified travel recommendations.

Overview...

Week 4 introduces multi-agent systems, where multiple specialized AI agents collaborate to solve complex problems more effectively than a single agent.

You will begin by understanding what multi-agent systems are, why they are useful, and when they should be used instead of single-agent architectures.

The session explores common agent collaboration workflows and introduces popular frameworks used to build multi-agent applications.

You will then learn how LangGraph enables structured multi-agent workflows through concepts such as state management, persistence, and memory stores.

By defining states and nodes, developers can construct graphs that coordinate agent interactions and control how information flows across the system.

The session also explains subagents and how they can be used to modularize complex tasks while avoiding unnecessary system complexity.

Finally, you will implement a practical project: a smart travel planning multi-agent system. In this system, specialized agents collaborate to search flights, find hotels, plan itineraries, answer travel queries, and generate personalized travel recommendations through coordinated decision making.

Note:

Post-class resources are optional and intended for post-class learning.

HOW TO START?

Roadmap

Pre-class Setup

- Tavily API Key
- Serp API Key

Live Class

Multi-Agent Systems Basics

- What, why, when
- Agent workflows & frameworks

LangGraph Foundations

- State, persistence, memory
- Build a simple graph (states + nodes)

Subagents Design

- When to use subagents
- When not to use them

Assignment

- Build a multi-agent system using LangGraph that acts as a smart content assistant

Post-class resources (Optional deep dives)

- Multi-Agent Systems & Orchestration

Learners are encouraged to follow along with the instructor during the live session and execute the code on their local environment.

Please use the Wednesday Technical Coaching session to clarify any doubts you have on the topics. The assignment solutions will be discussed during the Thursday Assignment Review session.

All the resources have been added to your Uplevel account. If you have any questions, please reach out to the Operations team or raise a support ticket.